

CO2xN: Porewater Methane

Month Year Samples

2026-06-05

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##Run Information

```
cat("Run Information: Melanie Giessner ") #lets you know what section you're in
```

Run Information: Melanie Giessner

```
#set the run date & user name
```

```
sample_year <- "2026"
```

```
user <- "Melanie Giessner"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month
```

```
folder_path_rawdata <- paste0("Raw Data")
```

```
GHG_files <- list.files(path = folder_path_rawdata,pattern = "STEEP" ,full.names = TRUE)
```

```
head(GHG_files)
```

```
## [1] "Raw Data/20260420_STEEP_GCReW.csv"
```

```
## [2] "Raw Data/20260504_STEEP_GCReW_SWH.csv"
```

```
## [3] "Raw Data/20260505_STEEP_SWH_MSM.csv"
```

```
## [4] "Raw Data/20260511_STEEP_MSM.csv"
```

```
## [5] "Raw Data/20260512_STEEP_GWI.csv"
```

```
## [6] "Raw Data/20260513_GENX_GHGdata_STEEP_GWI.csv"
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
log_path <- "Processed Data/ShimadzuGC_Slope_QAQC.csv"
```

```
# Define final path for data be sure to change year and month and run number
```

```
final_path <- "Processed Data/STEEP_N20_Data.csv"
```

```
#Gas in Water Calculations
```

```
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
```

```
TempC = 25 #Equilibration Temperature (Celsius)
```

```
volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
```

```
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g)
```

```
#record any notes about the run or anything other info here:
```

```
run_notes <- ""
```

```
#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"
```

```
cat(run_notes)
```

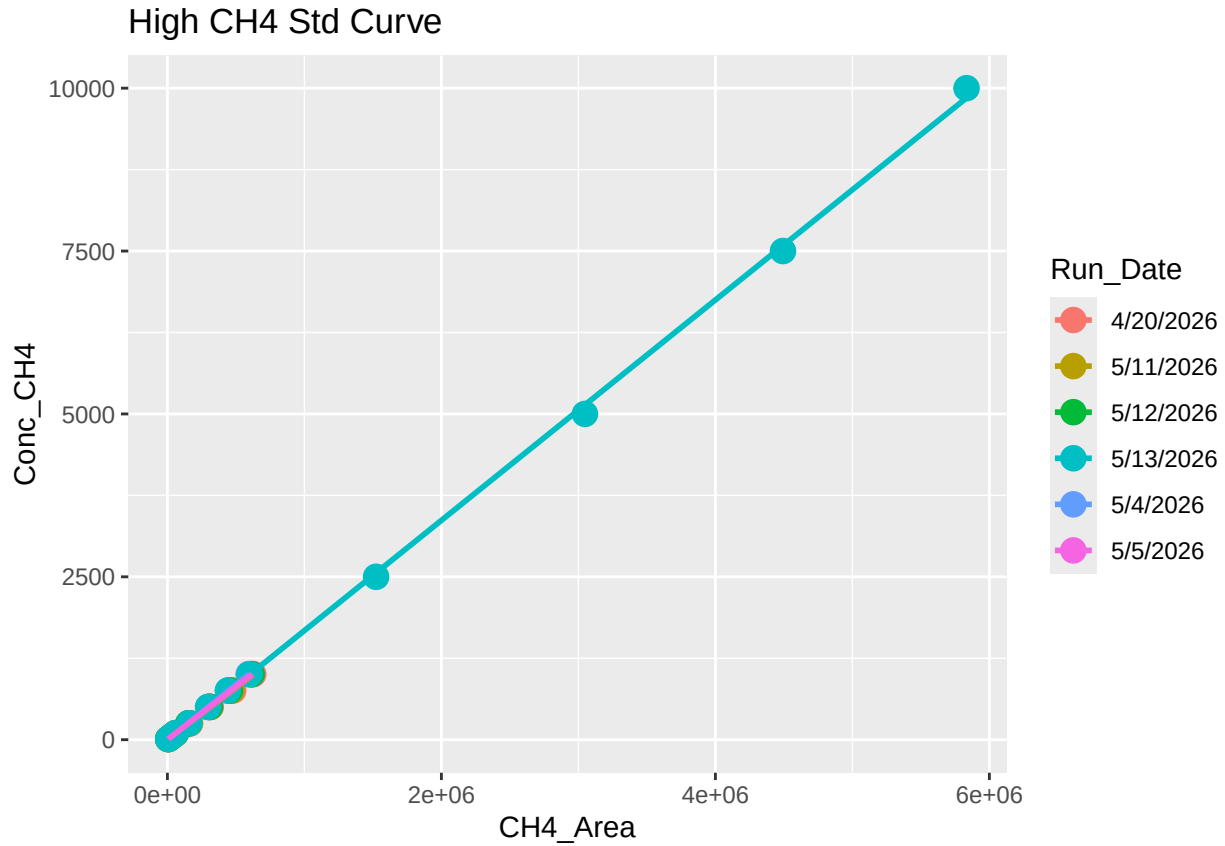
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

1 Read in data files

2 Assess standard curves

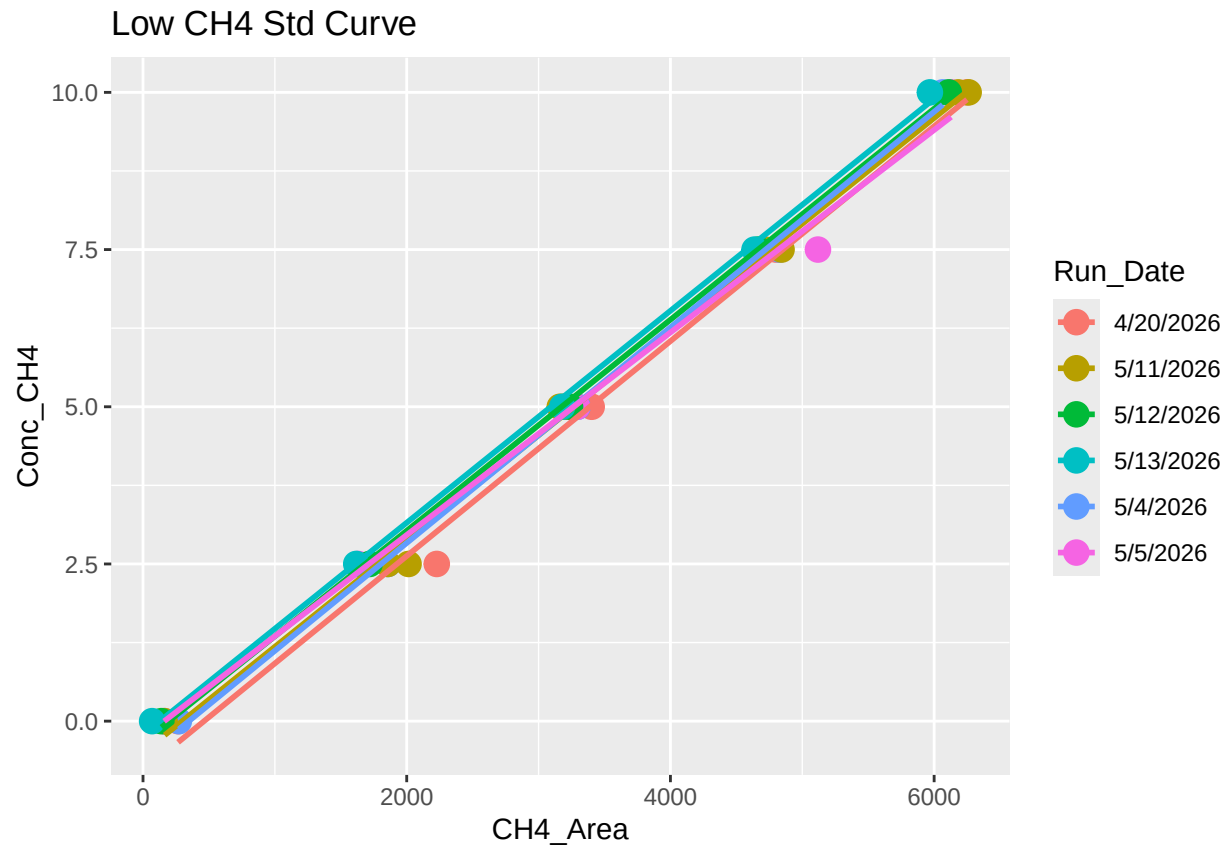
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```

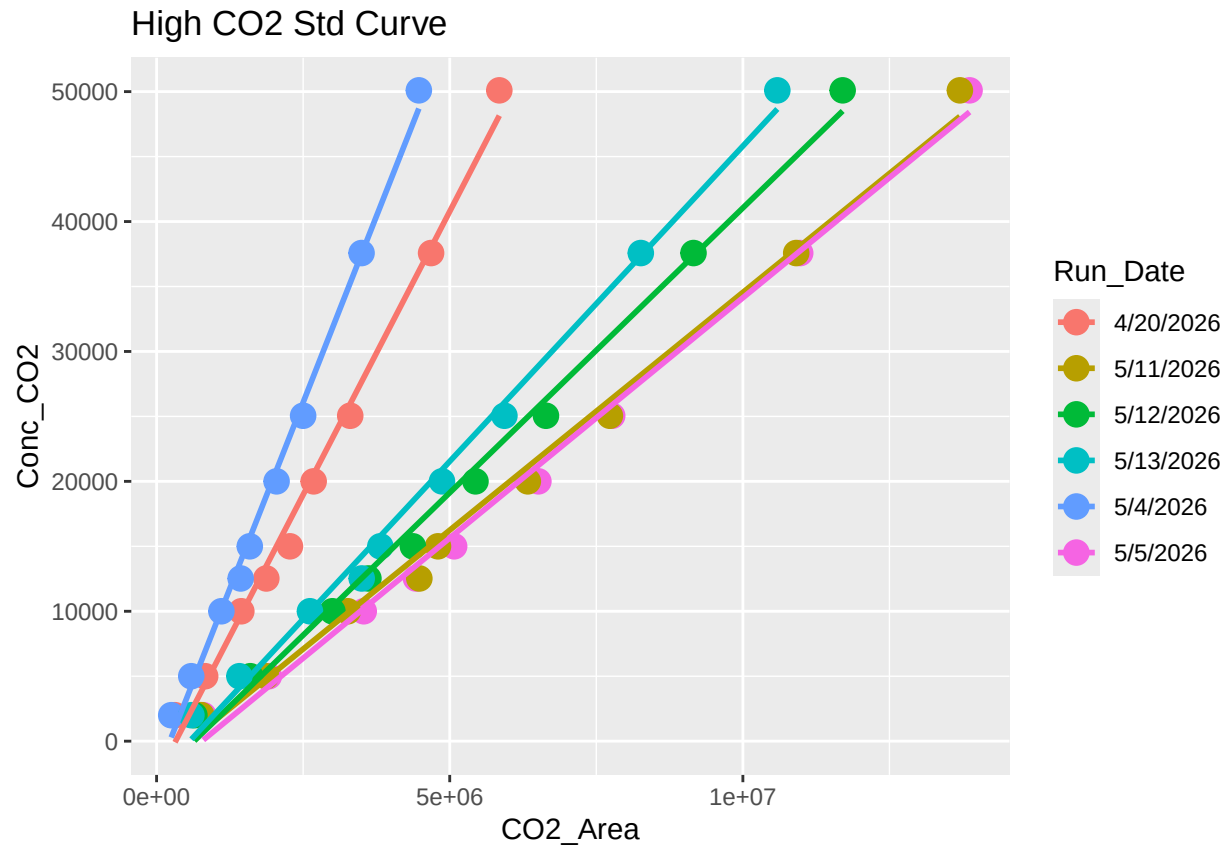


```
## Joining with 'by = join_by(Run_Date)'

## [1] "High_R2 is Good-4/20/2026" "High_R2 is Good-5/11/2026"
## [3] "High_R2 is Good-5/12/2026" "High_R2 is Good-5/13/2026"
## [5] "High_R2 is Good-5/4/2026"  "High_R2 is Good-5/5/2026"

## [1] "Low_R2 is Good-4/20/2026" "Low_R2 is Good-5/11/2026"
## [3] "Low_R2 is Good-5/12/2026" "Low_R2 is Good-5/13/2026"
## [5] "Low_R2 is Good-5/4/2026"  "Low_R2 is Good-5/5/2026"

## 'geom_smooth()' using formula = 'y ~ x'
```



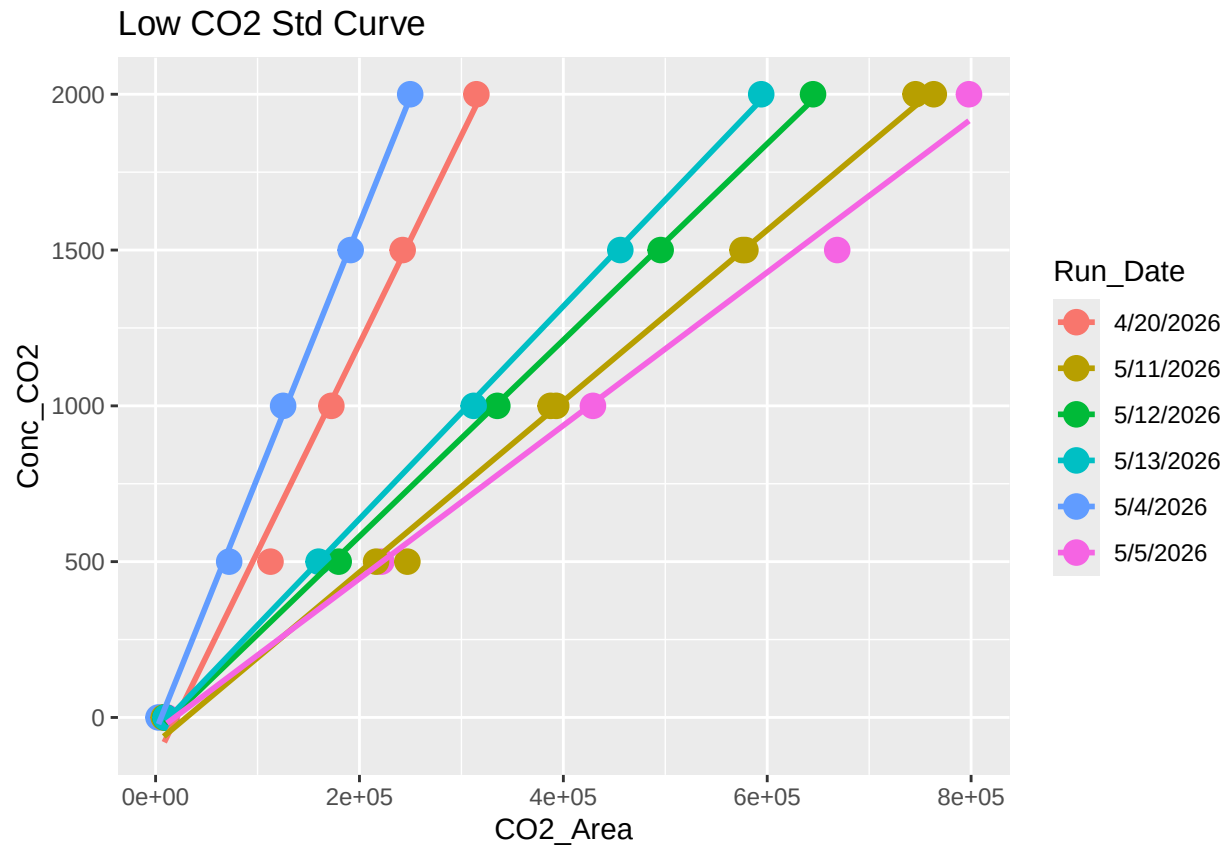
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
```

```
## ('stat_smooth()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
```

```
## ('geom_point()').
```



```
## Joining with 'by = join_by(Run_Date)'
```

```
## [1] "High_R2 is Good-4/20/2026" "High_R2 is Good-5/11/2026"
```

```
## [3] "High_R2 is Good-5/12/2026" "High_R2 is Good-5/13/2026"
```

```
## [5] "High_R2 is Good-5/4/2026" "High_R2 is Good-5/5/2026"
```

```
## [1] "Low_R2 is Bad-4/20/2026" "Low_R2 is Good-5/11/2026"
```

```
## [3] "Low_R2 is Good-5/12/2026" "Low_R2 is Good-5/13/2026"
```

```
## [5] "Low_R2 is Good-5/4/2026" "Low_R2 is Good-5/5/2026"
```

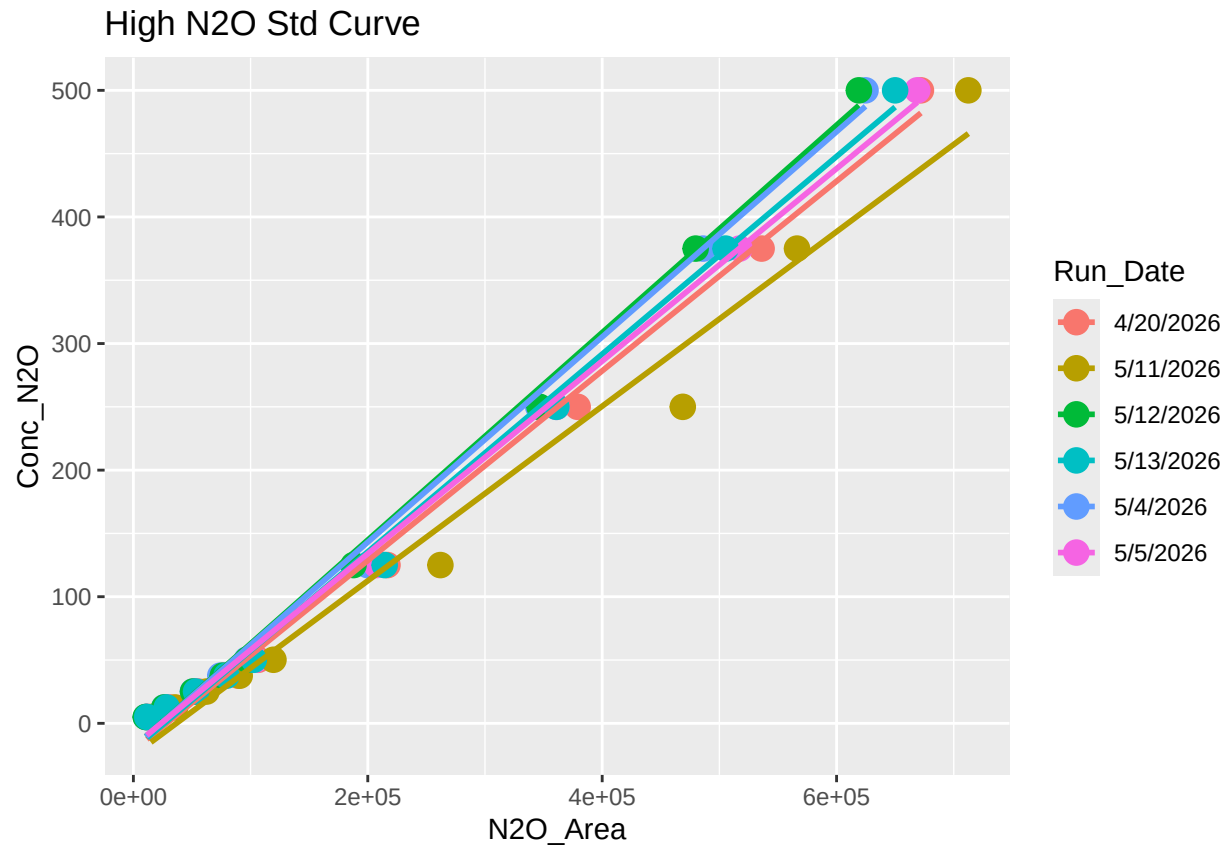
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
```

```
## ('stat_smooth()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
```

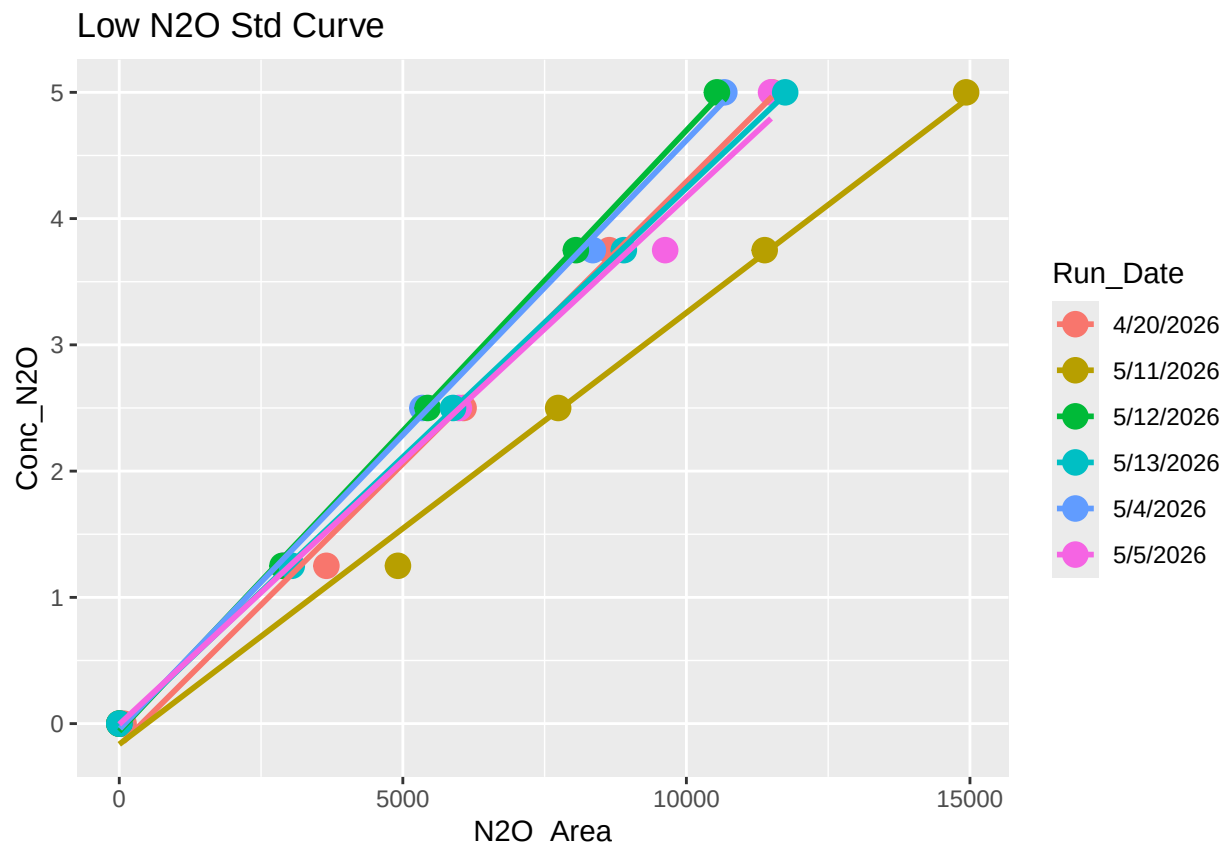
```
## ('geom_point()').
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 5 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

```
## Warning: Removed 5 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



```
## Joining with 'by = join_by(Run_Date)'
```

```
## [1] "High_R2 is Good-4/20/2026" "High_R2 is Bad-5/11/2026"
## [3] "High_R2 is Good-5/12/2026" "High_R2 is Good-5/13/2026"
## [5] "High_R2 is Good-5/4/2026" "High_R2 is Good-5/5/2026"
```

```
## [1] "Low_R2 is Good-4/20/2026" "Low_R2 is Good-5/11/2026"
## [3] "Low_R2 is Good-5/12/2026" "Low_R2 is Good-5/13/2026"
## [5] "Low_R2 is Good-5/4/2026" "Low_R2 is Good-5/5/2026"
```

```
#add new standard curve data to log
```

```
Slope_CH4_1 <- Slope_CH4 %>% mutate(curve = paste0("Slope_CH4_", CH4_Curve), R2 = CH4_R2, slope = CH4_S,
  CH4_Curve == "Low" ~ paste(CH4_low_std_crv_cutoff,"ppm"),
  CH4_Curve == "High" ~ paste(CH4_high_std_crv_cutoff,"ppm"),
  .default = as.character(CH4_mid_std_crv_cutoff)
)) %>% select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)
Slope_CO2_1 <- Slope_CO2 %>% mutate(curve = paste0("Slope_CO2_", CO2_Curve), R2 = CO2_R2, slope = CO2_S,
  CO2_Curve == "Low" ~ paste(CO2_low_std_crv_cutoff,"ppm"),
  CO2_Curve == "High" ~ paste(CO2_high_std_crv_cutoff,"ppm"),
  .default = as.character(CO2_mid_std_crv_cutoff)
)) %>% select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)
Slope_N2O_1 <- Slope_N2O %>% mutate(curve = paste0("Slope_N2O_", N2O_Curve), R2 = N2O_R2, slope = N2O_S,
  N2O_Curve == "Low" ~ paste(N2O_low_std_crv_cutoff,"ppm"),
  N2O_Curve == "High" ~ paste(N2O_high_std_crv_cutoff,"ppm"),
  .default = as.character(N2O_mid_std_crv_cutoff))
```



```

)) %>% select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)

Slope <- rbind(Slope_CH4_1, Slope_CO2_1, Slope_N2O_1)

log <- read.csv(log_path)
log <- log[, -c(1)]
head(log)

```

```

##   Analysis   Run_Date   Run_by Curve_top      curve      slope
## 1    CH4 2024-07-09 Alia Al-Haj    10 ppm Slope_Ch4_low 0.001594508
## 2    CH4 2024-07-09 Alia Al-Haj   100 ppm Slope_Ch4_high 0.001562326
## 3    N2O 2024-07-09 Alia Al-Haj    5 ppm Slope_N2O_low 0.000192023
## 4    N2O 2024-07-09 Alia Al-Haj  50.2 ppm Slope_N2O_high 0.000282245
## 5    CO2 2024-07-09 Alia Al-Haj  2000 ppm Slope_CO2_low 0.000282245
## 6    CO2 2024-07-09 Alia Al-Haj 20000 ppm Slope_CO2_high 0.001549502
##   intercept      R2
## 1 -0.20901179 0.9960215
## 2  0.09646243 0.9996531
## 3  0.07848462 0.9940785
## 4 -0.49925490 0.9960644
## 5 -0.49925490      NA
## 6 38.64567509 0.9996907

```

```

#log <-log %>%mutate(Run_Date = as.Date(Run_Date,"%m/%d/%Y"))
log <- rbind(log, Slope)

#remove possible duplicates from multiple QAQC attempts

unique_log <- log %>% distinct(.keep_all = TRUE)%>%
  mutate(Run_Date = as.Date(Run_Date,"%m/%d/%Y"), curve= gsub("L","l",curve), curve= gsub("H","h",c

Slopes_chk_CH4 <- ggplot(unique_log %>% filter( grepl("CH4",curve,ignore.case = T)), aes(x = Run_Date,
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()

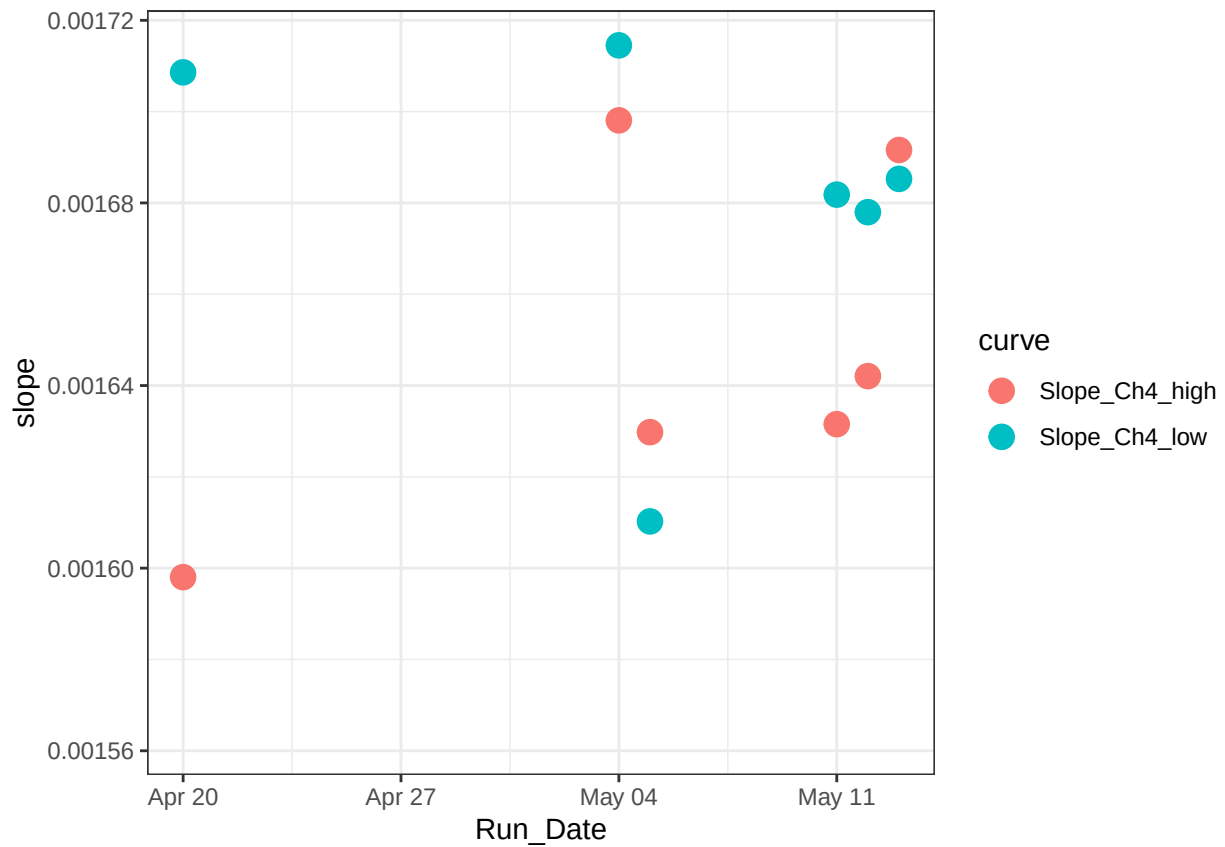
Slopes_chk_CH4

```

```

## Warning: Removed 34 rows containing missing values or values outside the scale range
## ('geom_point()').

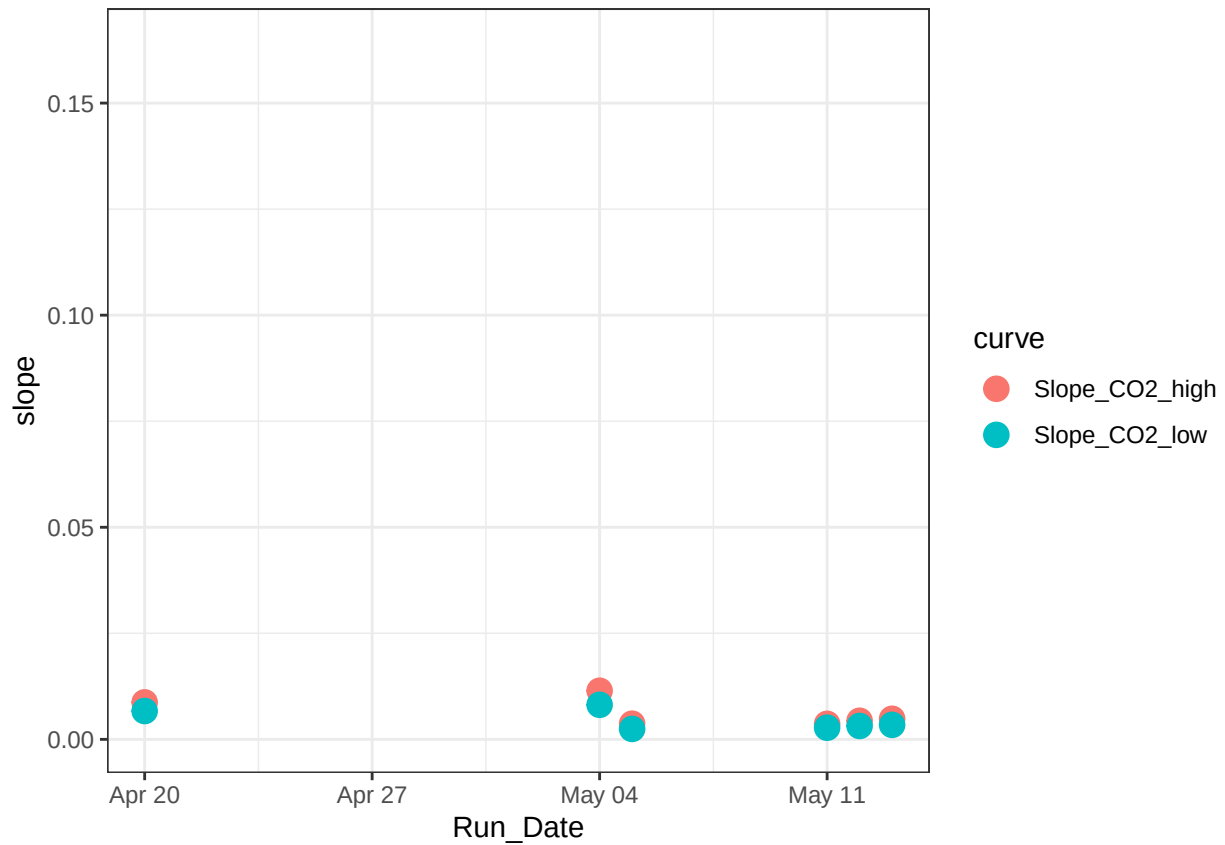
```



```
Slopes_chk_C02 <- ggplot(unique_log %>% filter( grepl("C02",curve,ignore.case = T)), aes(x = Run_Date,
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

Slopes_chk_C02

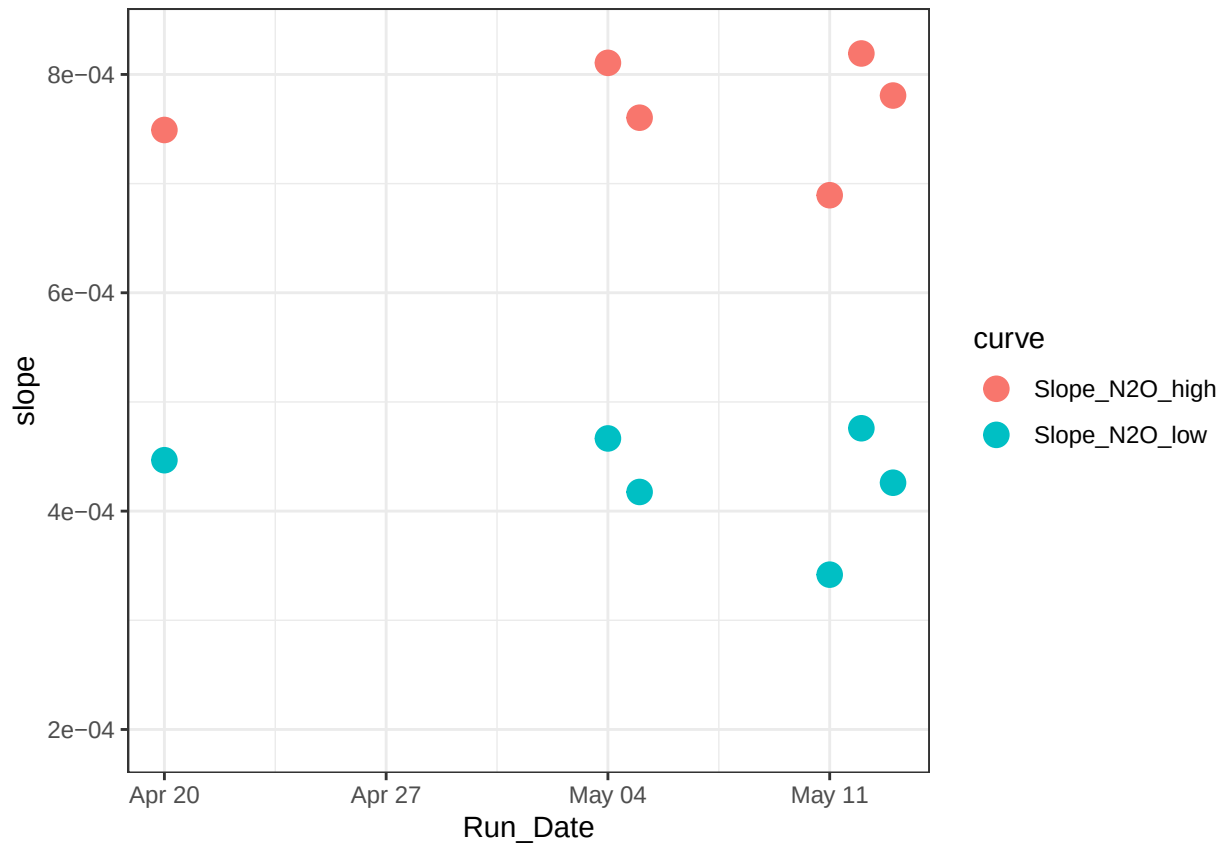
```
## Warning: Removed 35 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
Slopes_chk_N20 <- ggplot(unique_log %>% filter( grepl("N20",curve,ignore.case = T)), aes(x = Run_Date,
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

Slopes_chk_N20

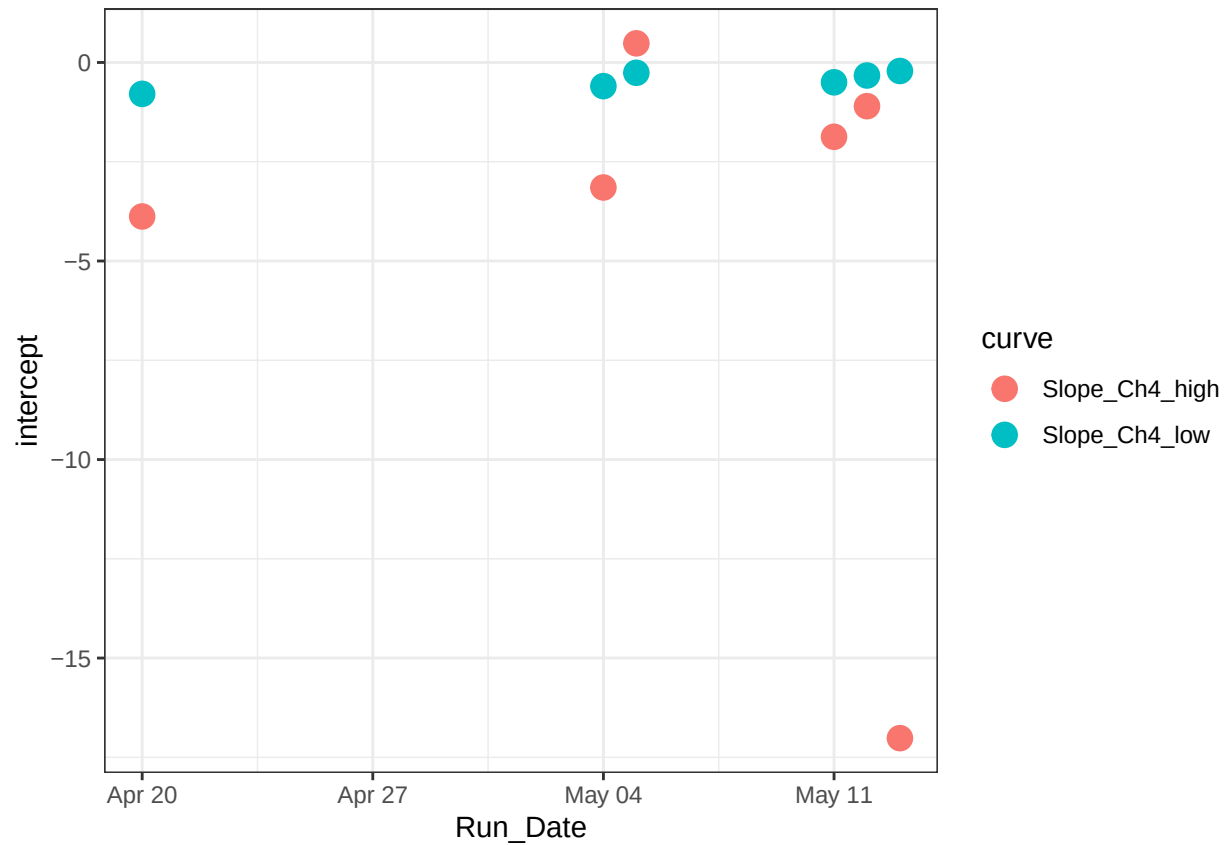
```
## Warning: Removed 34 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
int_chk_CH4 <- ggplot(unique_log %>% filter( grepl("CH4",curve,ignore.case = T)), aes(x = Run_Date, y = slope)) +
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

```
int_chk_CH4
```

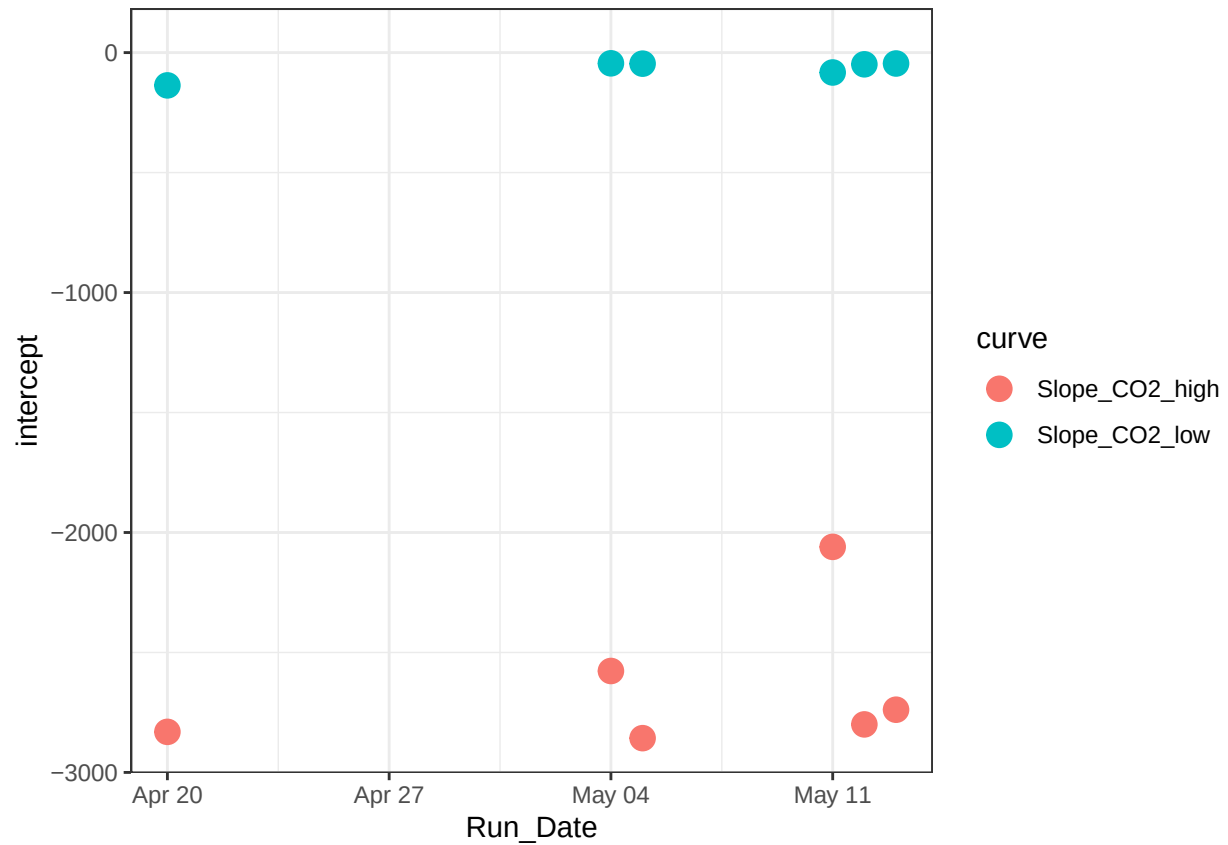
```
## Warning: Removed 34 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
int_chk_C02 <- ggplot(unique_log %>% filter( grepl("C02",curve,ignore.case = T)), aes(x = Run_Date, y = intercept)) +
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

```
int_chk_C02
```

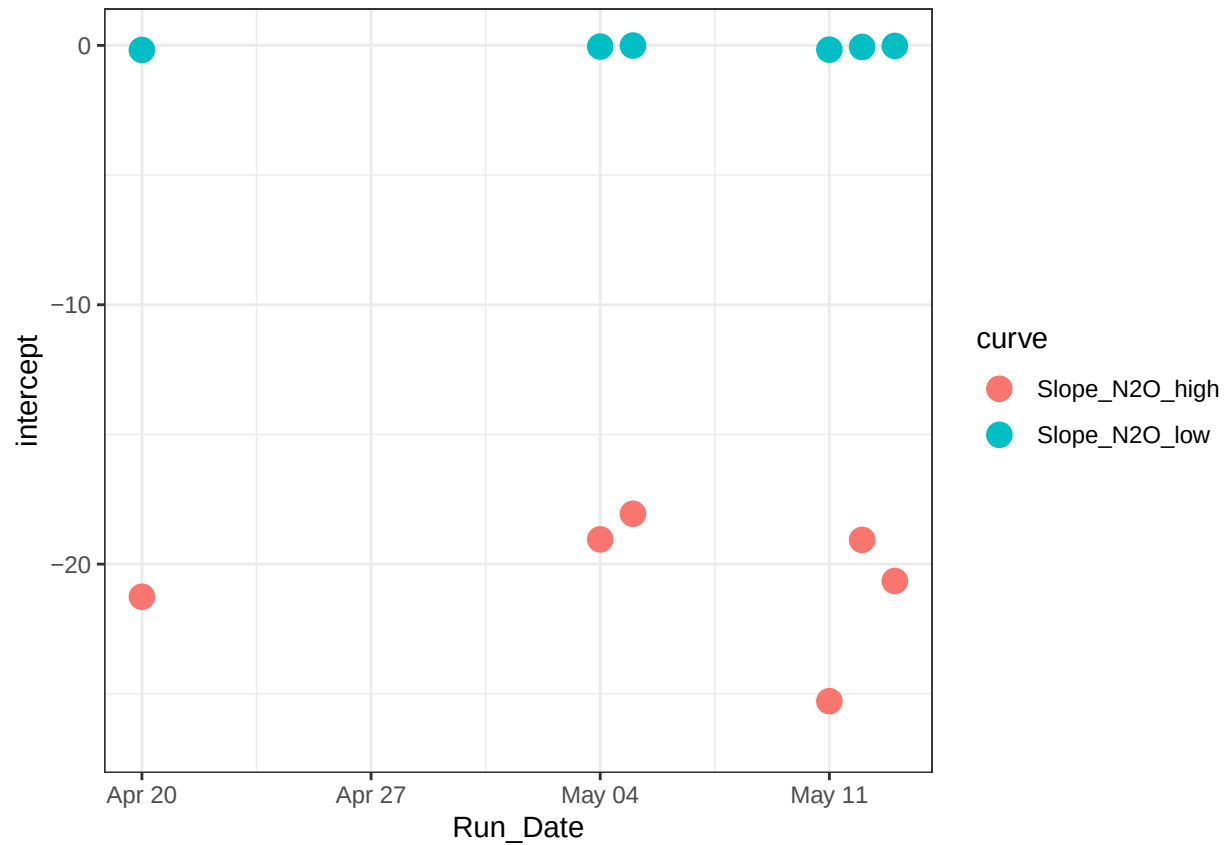
```
## Warning: Removed 35 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
int_chk_N20 <- ggplot(unique_log %>% filter( grepl("N20",curve,ignore.case = T)), aes(x = Run_Date, y = intercept)) +
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

```
int_chk_N20
```

```
## Warning: Removed 34 rows containing missing values or values outside the scale range
## ('geom_point()').
```



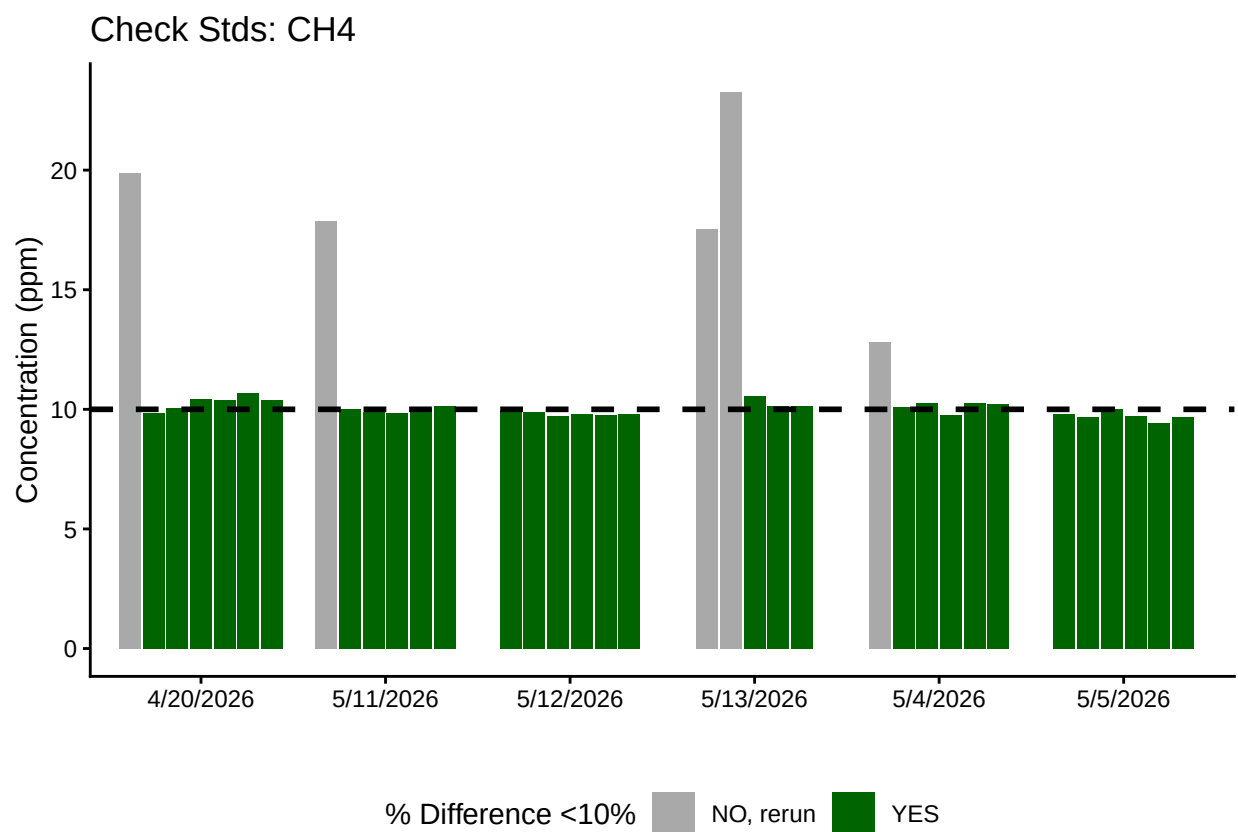
2.1 Now calculate the CH₄ in ppm

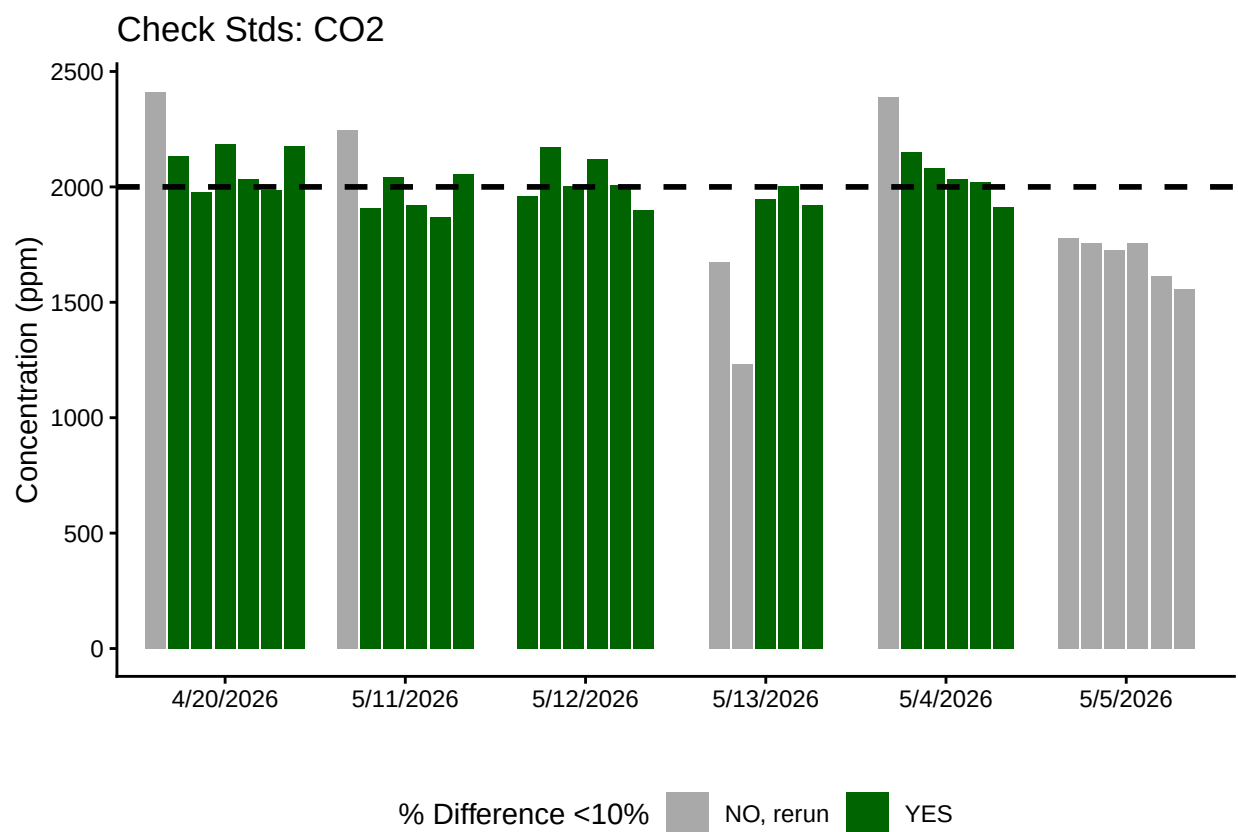
3 Aanlayze the Check Standards

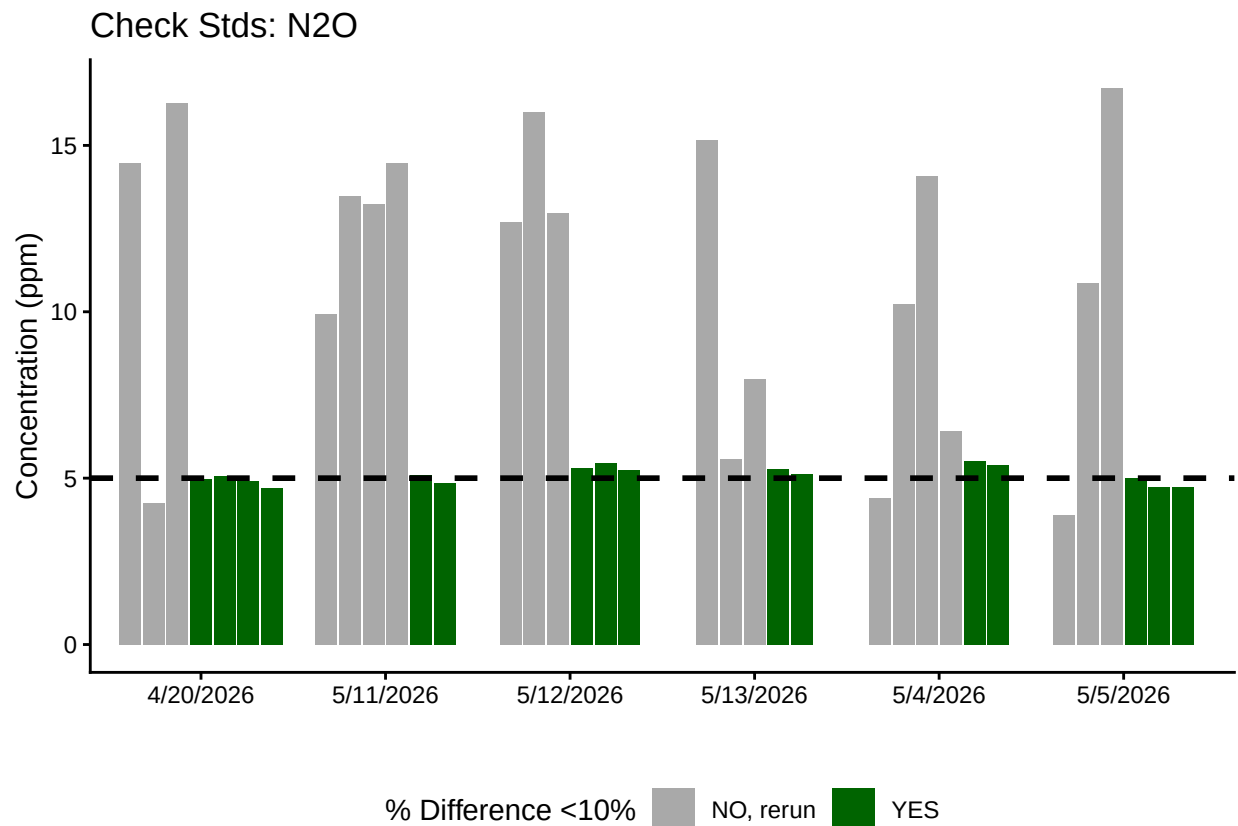
>60% of CH₄ Check Standards are within range of expected concentration - PROCEED

>60% of CO₂ Check Standards are within range of expected concentration - PROCEED

<60% of N₂O Check Standards are within range of expected concentration - REASSESS







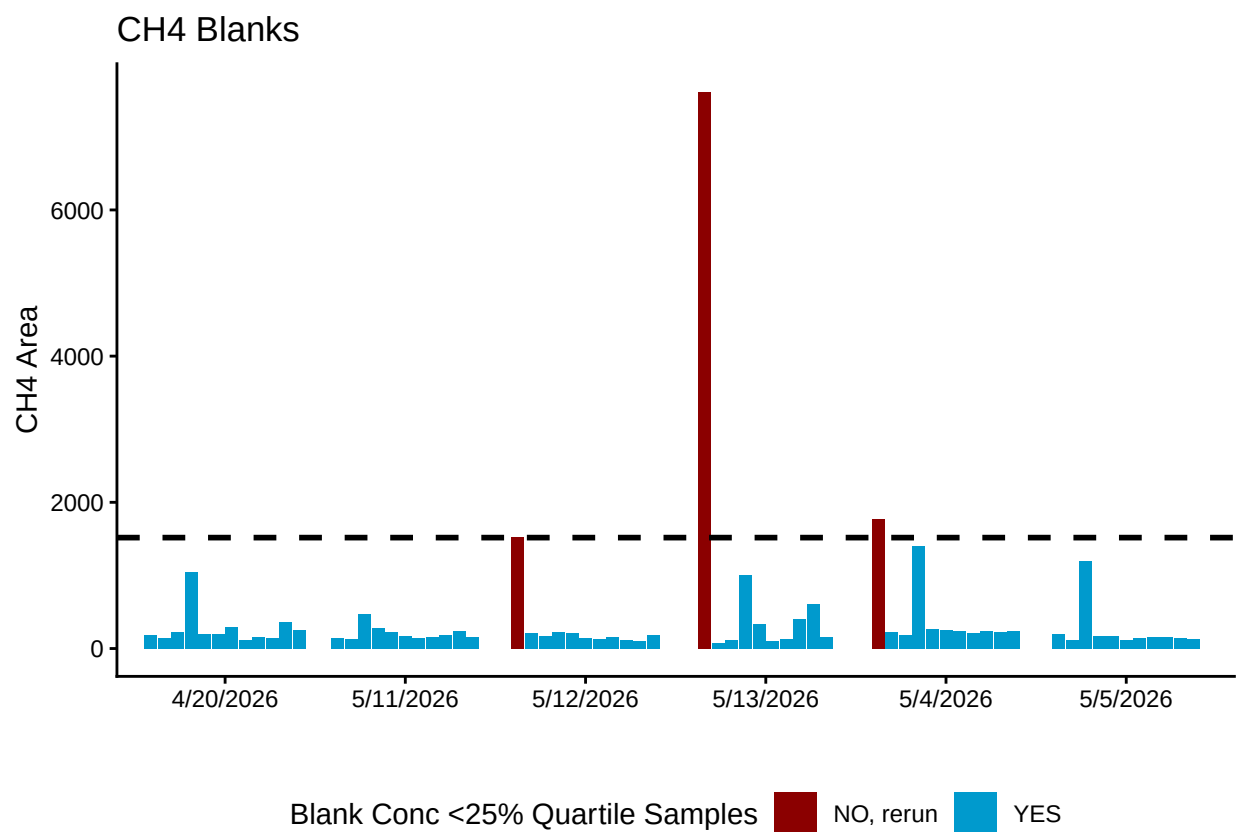
4 Analyze the Blanks

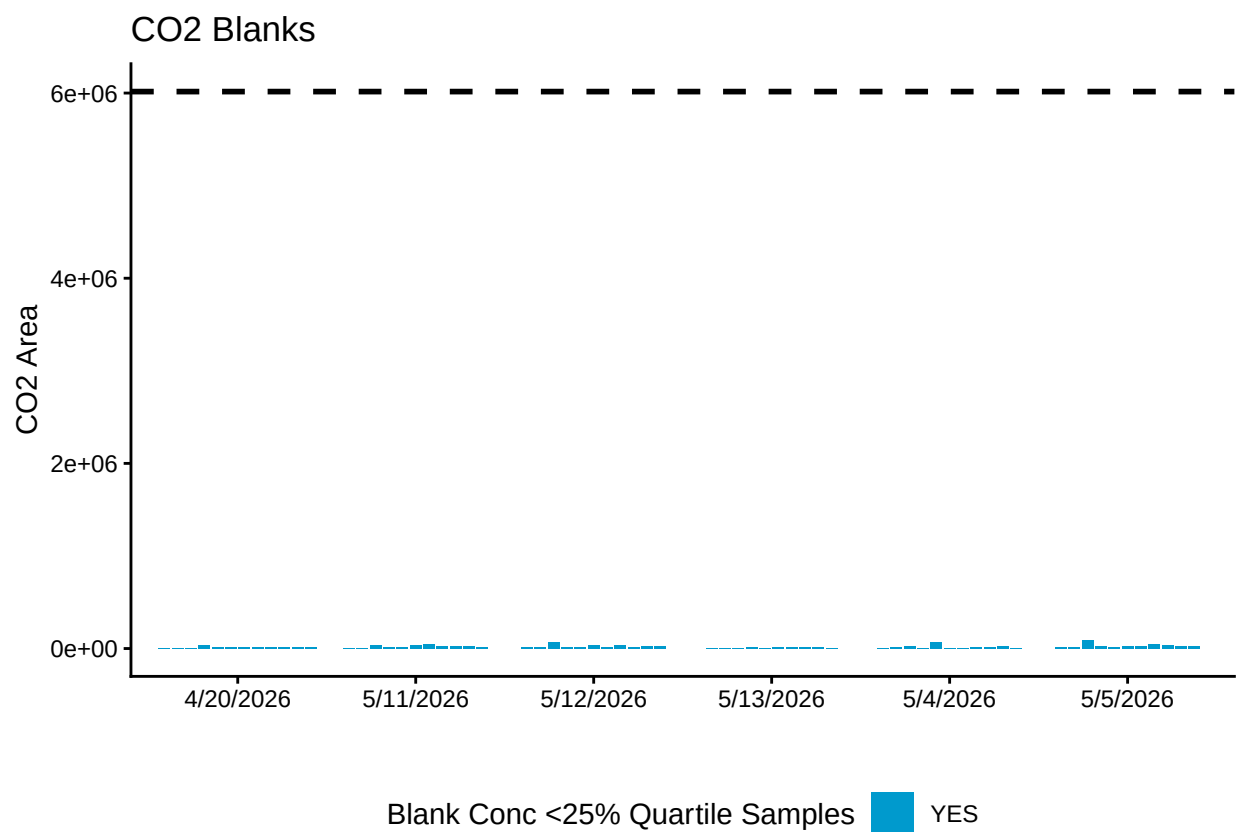
Assess Blanks

>60% of CH₄ Blanks are below the lower 25% quartile of samples - PROCEED

>60% of CO₂ Blanks are below the lower 25% quartile of samples - PROCEED

>60% of N₂O Blanks are below the lower 25% quartile of samples - PROCEED





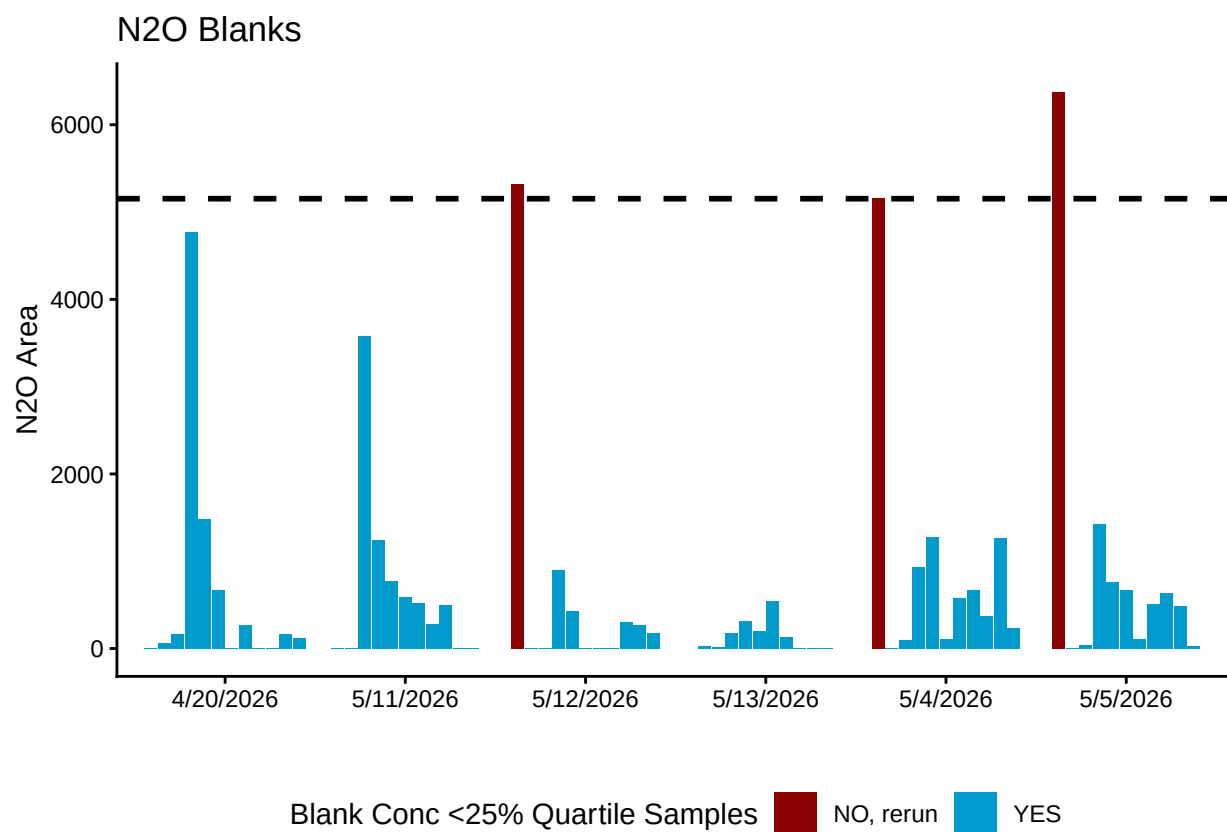
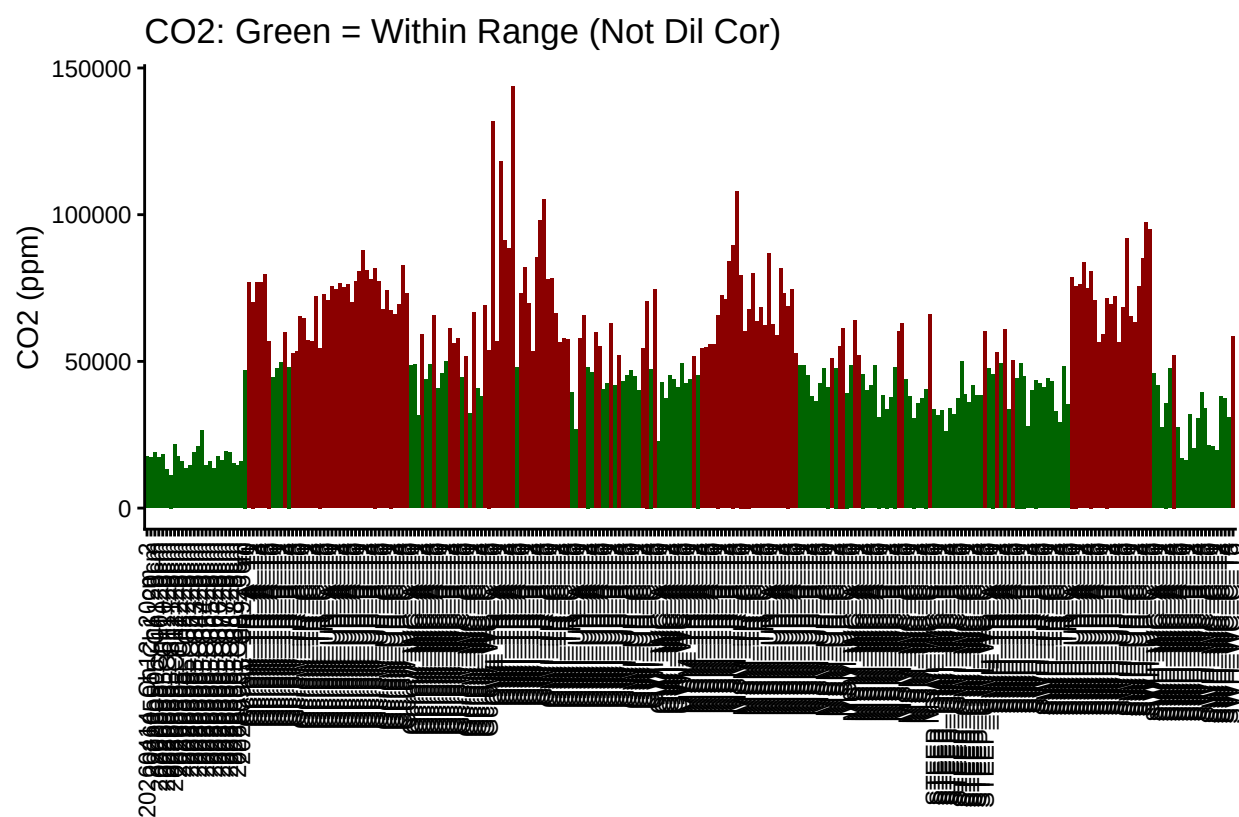
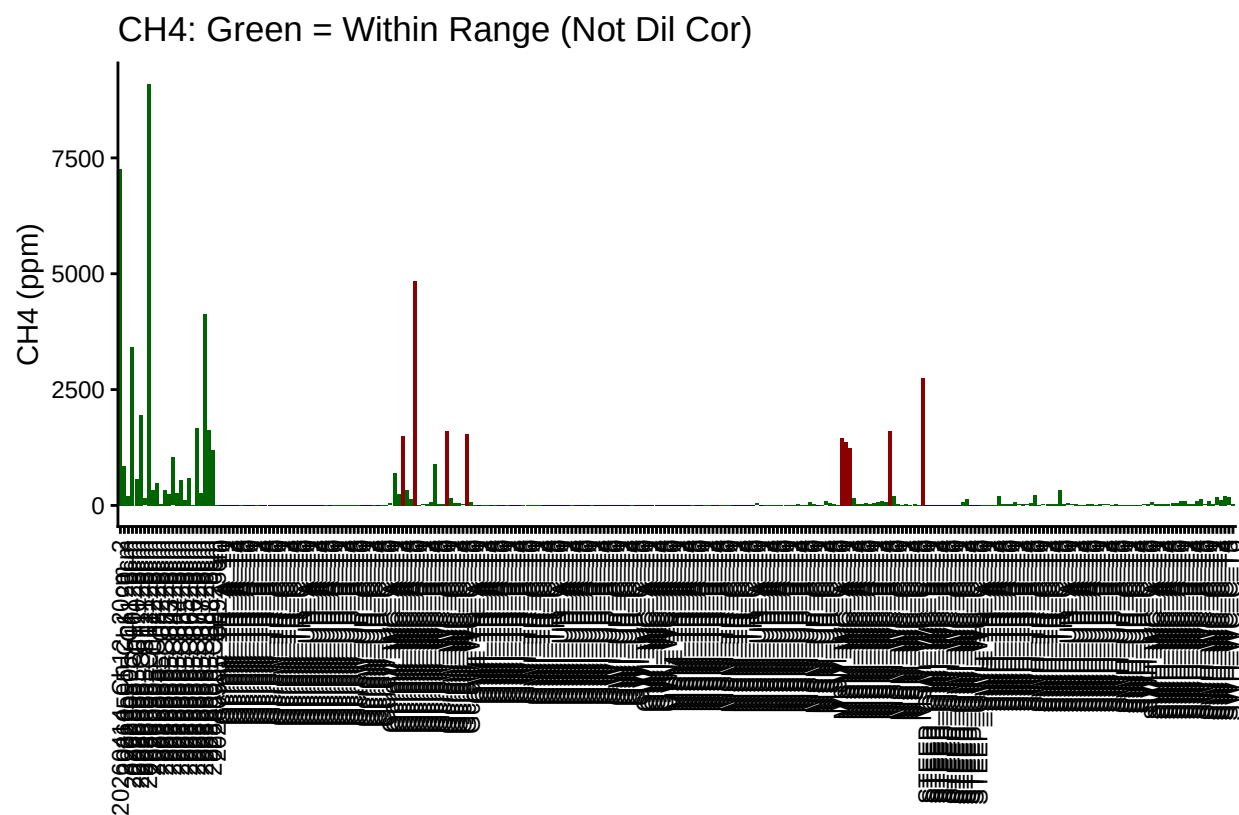


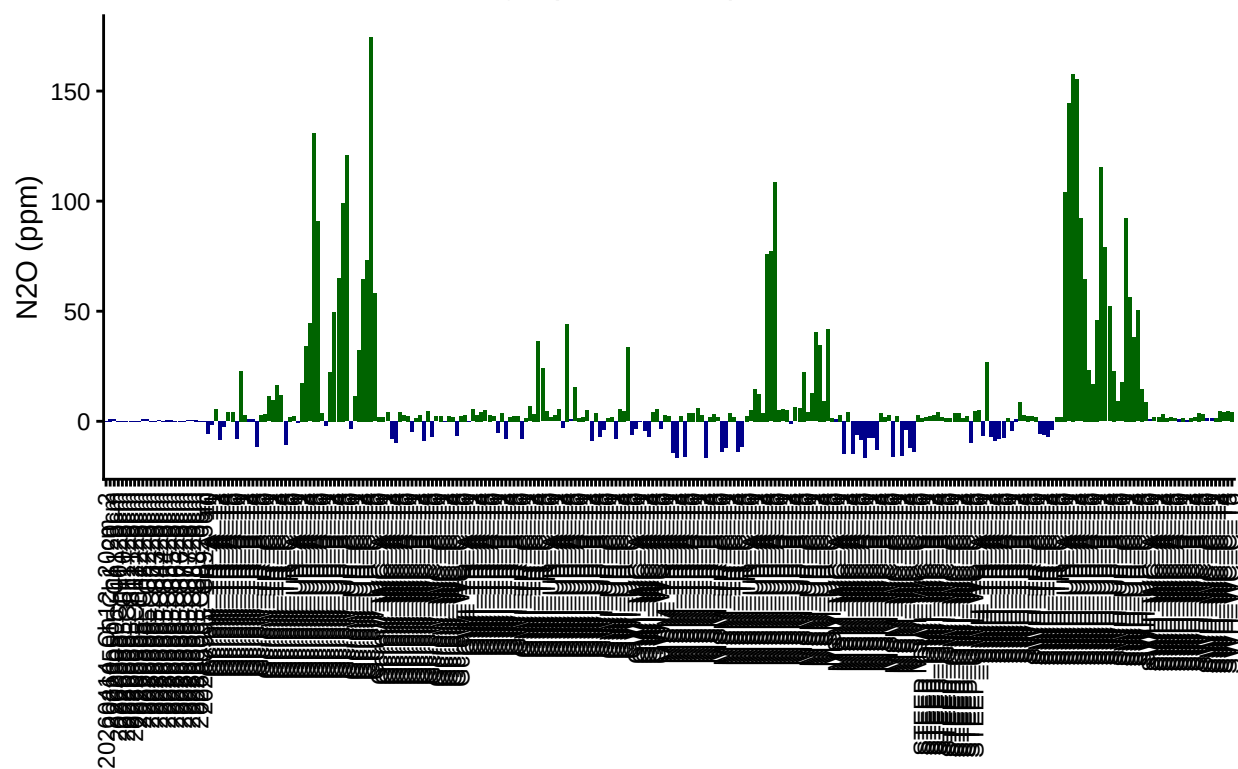
Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	414.2121
CO2	22032.2576
N2O	693.3485

5 Dilution correct samples and Check Range

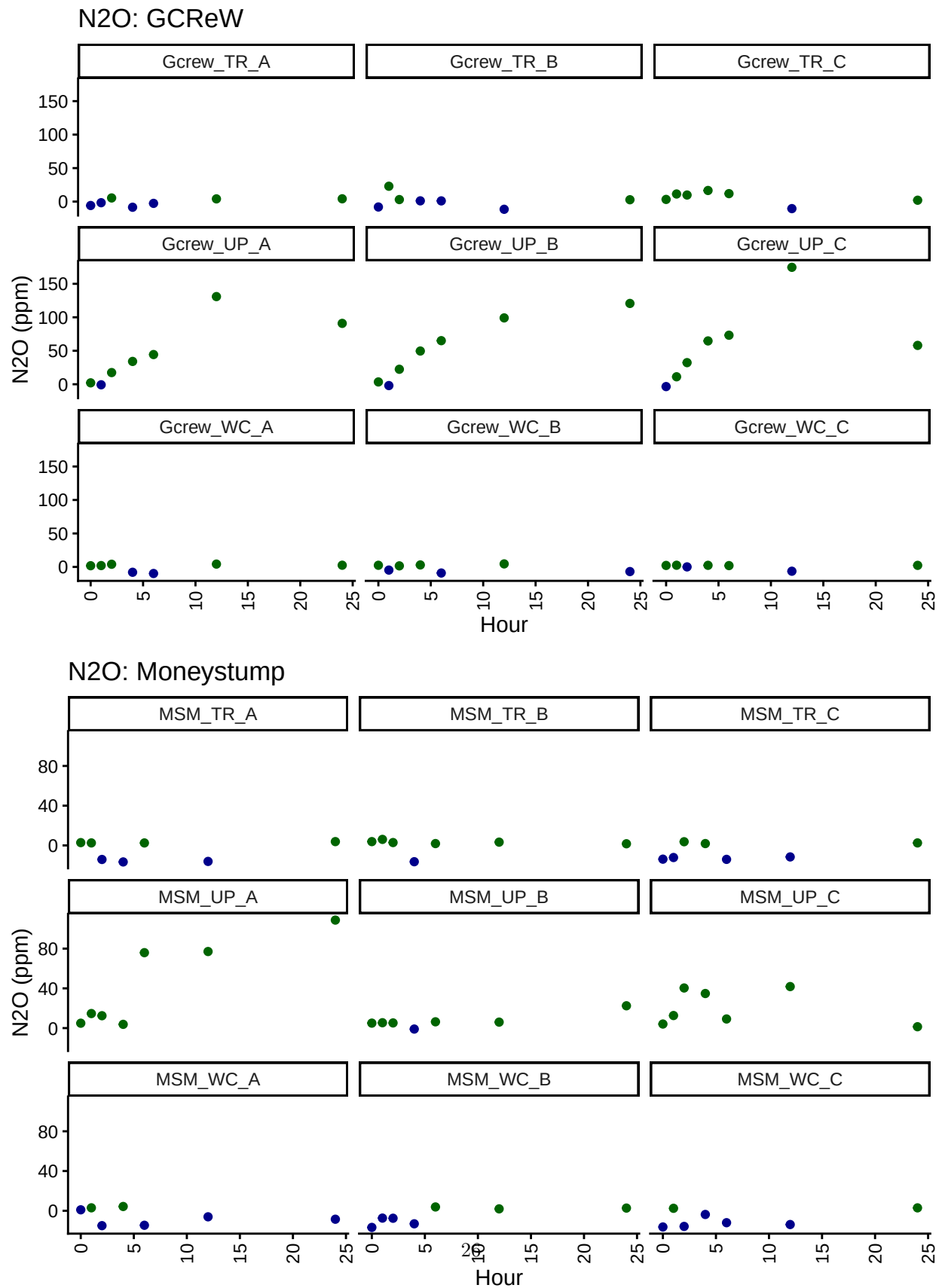


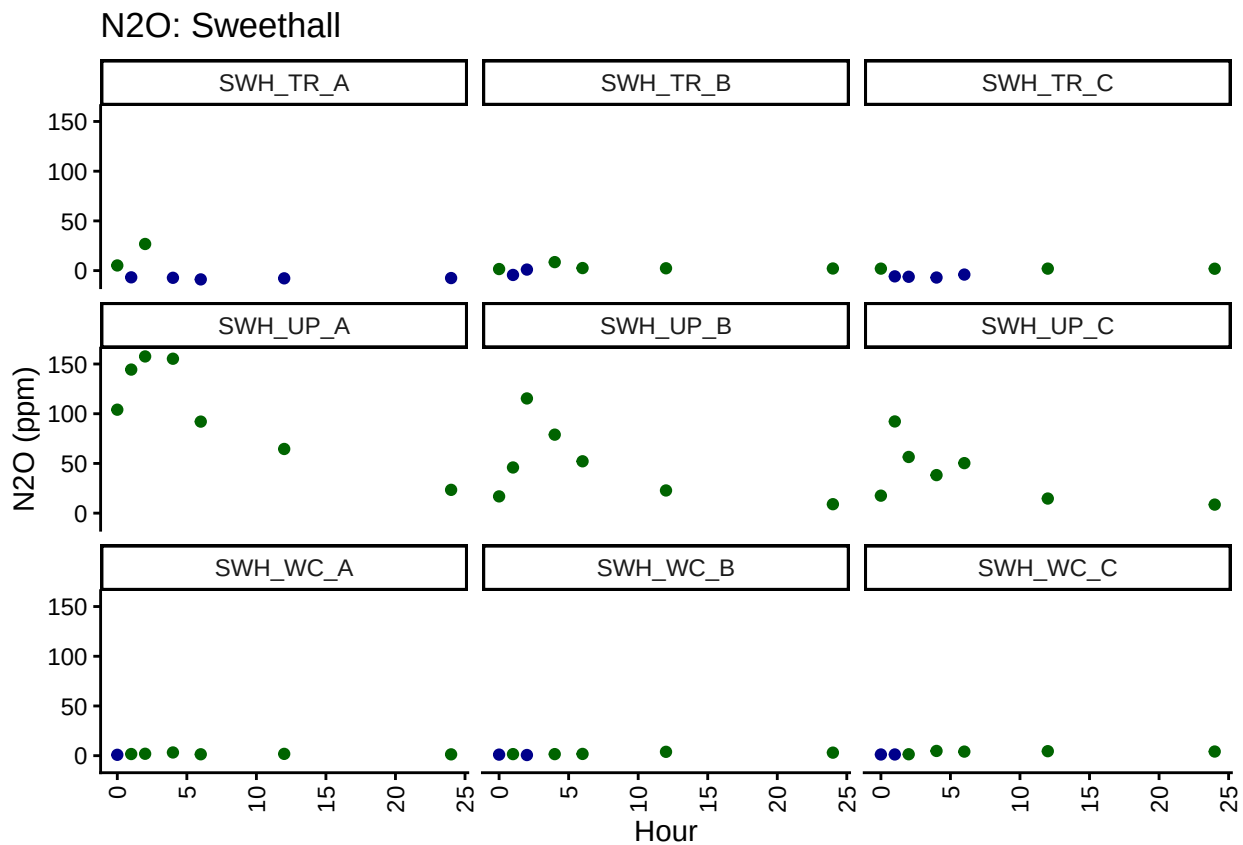
N2O: Green = Within Range (Not Dil Cor)

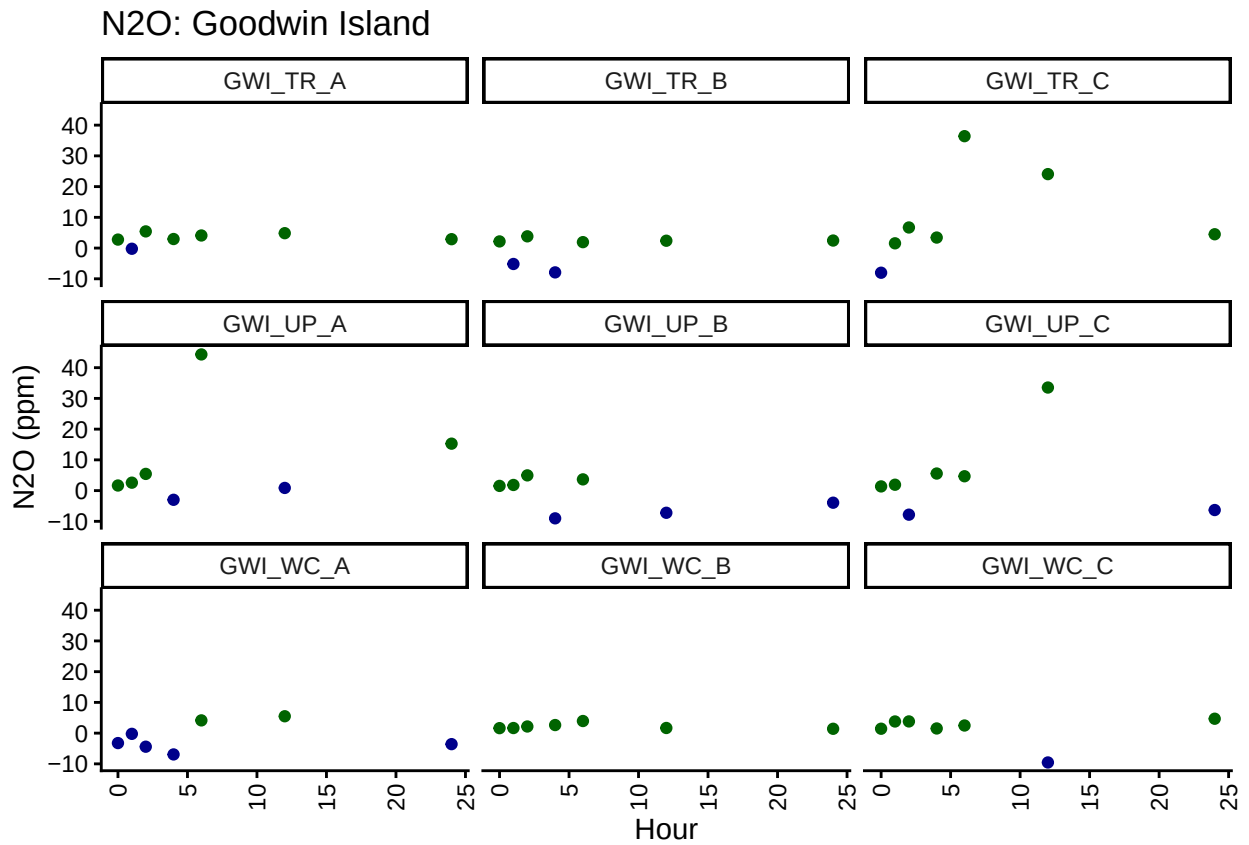


6 Calculate gas in water

6.1 Visualize Data







##Export Data

#end